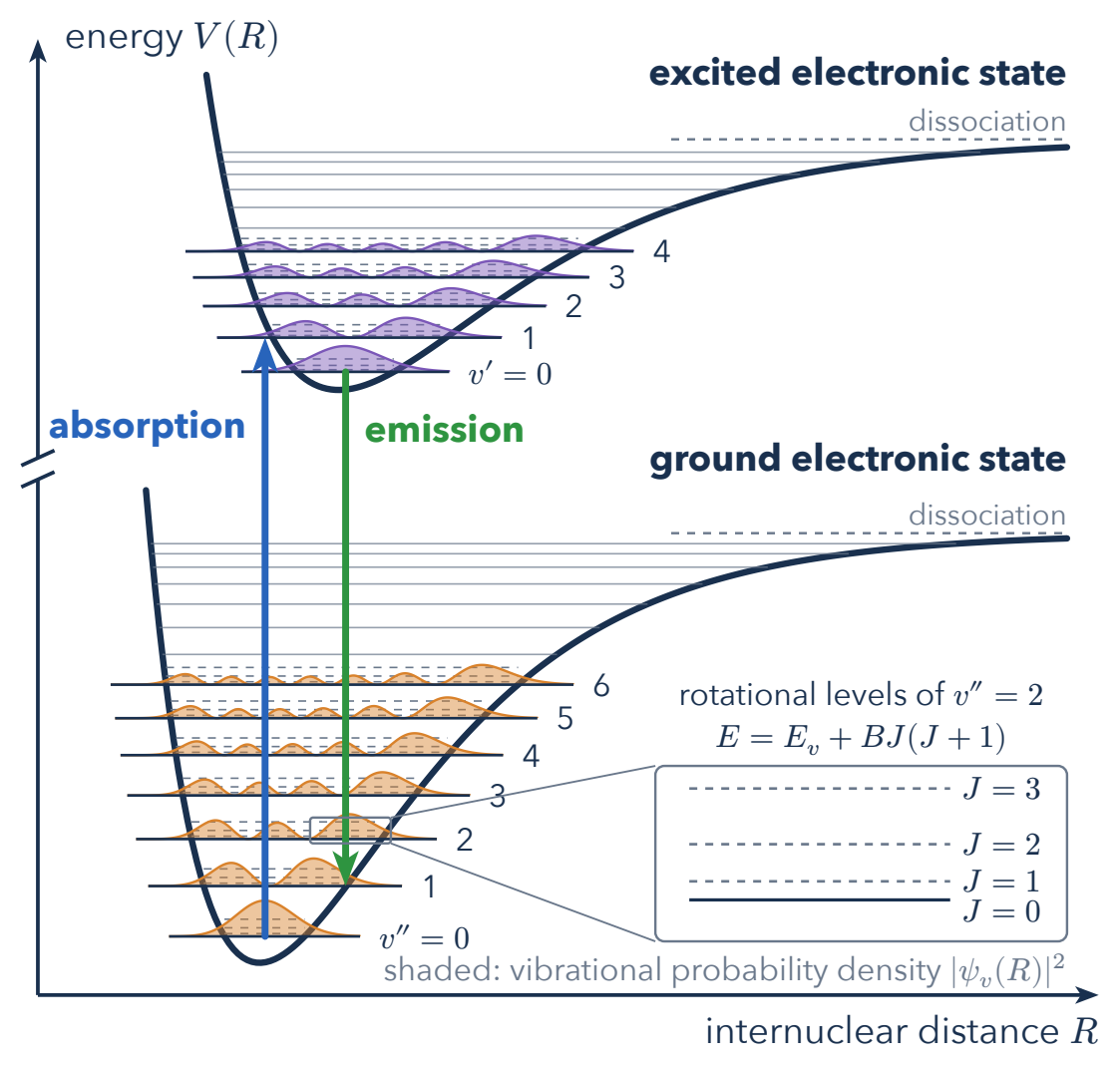

energy $V(R)$
excited electronic state
dissociation
4
3
2
1
$v' = 0$
absorption
emission
ground electronic state
dissociation
6
5
4
3
2
1
$v'' = 0$
rotational levels of $v'' = 2$
$E = E_v + BJ(J+1)$
$J = 3$
$J = 2$
$J = 1$
$J = 0$
shaded: vibrational probability density $|\psi_v(R)|^2$
internuclear distance R